

accident. Vehicle-to-infrastructure (V2I) technology is a communication structure that enables many vehicles to share data with a range of devices supporting the highway system of a given country.

These devices include RFID readers, signage, cameras, lane makers, streetlights, parking meters, and more. Enabled by a network of hardware, software, and firmware, the V2I technology is usually wireless and bi-directional. It enables vehicles to communicate with infrastructure, such as road signs, traffic signals, and traffic management systems (TMS), to provide information about safety issues and the best possible route to take based on the prevailing traffic conditions, accidents, or breakages.

V2I also allows autonomous parking. A garage or parking area 'talks' to the vehicle via the Internet and informs about the free space. After dropping off passengers, the vehicle approaches the free space and automatically parks itself. To get the vehicle back from the parking, the driver has to just make a call using his/her smartphone. A recent study by the National Highway Traffic Safety Administration (NHTSA) has reported that V2V systems, together with V2I, could reduce road accidents by about eighty per cent.

Augmented reality dashboards

Augmented reality (AR) display can overlay information on top of what a driver is actually seeing in real life. BMW has already implemented a windshield display in some of their vehicles, which displays basic information. They're also developing augmented reality dashboards that will help identify any external object in front of a vehicle and inform the driver how far away he/she is from that object.

So, if a driver is approaching another vehicle too fast, a red sign will appear on the approaching vehicle's windshield, and directions will be displayed showing how to manoeuvre to avoid collision.

BMW is also researching the use of AR for technicians. Using AR glasses, technicians will be able to find which part of the vehicle needs to be serviced or replaced. The display will show the step-by-step instructions on how to

go about the entire process.

AR is also being researched to allow passengers to zoom in on objects outside the car and select, identify, and view the object using a touchscreen window.

Airbags help stop cars

Airbags will not only provide improved protection to passengers but also increase the stopping power of automobiles in future. Mercedes is working to make airbags part of its vehicle safety system (VSS). The company is experimenting with airbags that deploy from underneath the car and help stop a vehicle before a crash.

These airbags are part of the overall active safety system and deploy when sensors determine that impact is inevitable. The bags have a friction coating that slows the car and can double the stopping power of the vehicle. The bags also lift the vehicle up to eight centimetres, which counters the car's dipping motion during hard braking, improves bumper-to-bumper contact, and helps prevent passengers from sliding under seat belts during a collision.

Energy-storing lightweight body panels

Manufacturers are researching and testing body panels made of polymer fibre and carbon resin that are strong, pliable, light in weight, store energy, and also charge lithium-ion batteries. These are capable of capturing energy produced by regenerative braking or when the vehicle is being charged overnight. Energy is then released as per the vehicle's demand.

Body panels are also being developed for capturing and storing solar energy. One of the major problems faced in designing hybrid vehicles is that batteries have a significant amount of weight and occupy a lot of space. This is where energy-storing lightweight body panels come to play.

Wireless technologies

Wireless technologies such as LTE, Wi-Fi, Bluetooth, and 5G allow vehi-



AR dashboards can identify external objects and display their information on the windshield (Credit: <https://technoupdate2017.wordpress.com>)

cles to communicate with infrastructure (V2I), network (V2N), people (V2P), and one another (V2V).

Continuous advancements are being made by automobile companies such as Toyota, Lexus, and Volkswagen and chip designing and manufacturing companies like Qualcomm and NXP. These advancements include technologies for faster data transmission, improved safety of vehicles, reduced latency of less than five milliseconds, and better data sharing.

Level-4 autonomous vehicles

As per the norms of Society of Automotive Engineers (SAE), level-0, level-1, and level-2 autonomy include advanced driver-assistance systems (ADAS) that have the capability of controlling vehicles autonomously in limited and brief conditions. These systems mainly include adaptive cruise control, parallel parking capability, and collision avoidance.

Level-3 or level-4 autonomy refers to driverless cars with autonomy and the ability to make decisions on their own. Companies like Waymo, Cruise, Uber, Lyft, Argo AI, and Voyage are approaching level-3 autonomy using a 'Safety Driver,' which is currently at the development stage. When level-4 vehicles would become a reality, accidents and traffic would get reduced drastically, and there would be no drunk driving incidents.

The problem being faced in reaching level 4 is that enough 'categorised data' is not available to teach vehicles how to drive in millions of different case scenarios. Another problem is the complete optimisation of algorithms to decide where the car should go based on the inputs from the available data.